

Rye Stahle-Smith

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EDUCATION

University of South Carolina

B.S.E. in Computer Engineering, Minor in Mathematics, GLD in Research

Honors: Dean's List - Fall 2023, Spring 2024, Spring 2025 & Fall 2025 | Tau Sigma NHS

Columbia, SC

Expected Graduation May 2026

GPA: 3.8

UNC at Asheville & North Carolina State University

Joint B.S. in Mechatronics Engineering

Honors: Dean's List - Fall 2021 | Order of Omega NHS | Morehead Family Engineering Scholarship

Asheville & Raleigh, NC

August 2021 – May 2023

RELATED EXPERIENCE

University of South Carolina | *Undergraduate Research Assistant* | Columbia, SC

May 2025 – Present

- Conducted R&D in reconfigurable embedded systems and cloud-scale computing, focusing on hardware security and real-time ML deployment.
- Developed an ML framework to detect hardware Trojans directly from FPGA bitstreams for efficient, real-time analysis.
- Primary maintainer of the open-source tool BLADEI; expanded support to all PYNQ-compatible FPGA platforms and transitioned the model pipeline from serialization to quantization, improving latency and portability.
- Managed concurrent research projects, balancing experimentation, documentation, and conference deadlines.
- Presented at the ACM Student Research Competition during The 38th International Conference for High-Performance Computing, Networking, Storage and Analysis (SC25).
- Accepted for publication and presentation at IEEE SoutheastCon 2026.

University of South Carolina | *McNair Junior Scholars Fellow* | Columbia, SC

May 2025 – August 2025

- Collaborated with research advisor and team to develop software and hardware skills.
- Used Vivado and Vitis to run hardware tests and support team projects.
- Conducted experiments on FPGA boards and compared performance.
- Completed an independent research project and presented at the 2025 Summer Research Symposium.

Outlier | *AI Training Specialist* | Remote, USA

October 2024 – October 2025

- Evaluated CS-related prompts to train generative AI models, improving their coding efficiency and accuracy.
- Ensured AI solutions met the standards of clarity, correctness, and creativity, including solving math problems and verifying API calls.
- Displayed proficiency in Java, Python, Verilog, and C++ while delivering precise attention to code quality, style, and grammar.

Personal Projects

BLADEI: Bitstream-Level Abnormality Detection for Embedded Inference

May 2025 – Present

- Re-engineered state-of-the-art (SOTA) hardware benchmarks from Trust-Hub, producing a labeled dataset of benign and malicious bitstreams.
- Built a supervised machine learning pipeline using byte-frequency analysis, dimensionality reduction with TSVD, and class balancing with SMOTE to identify malicious bitstreams.
- Deployed the final quantized model on PYNQ-supported FPGA devices to validate real-time performance in an embedded environment.

Three-Stage RISC-V Pipelined Processor

August 2024 – December 2024

- Designed and implemented a three-stage pipelined processor in Verilog, supporting a subset of the RV32I Base Integer Instruction Set for RISC-V, including R-, I-, U-, J-, and B-type instructions.
- Optimized the processor for efficient deployment on the DE2-115 FPGA board.

WORK EXPERIENCE

Motor Supply Company Bistro | *Server* | Columbia, SC

May 2024 – Present

- Gained in-depth familiarity with the restaurant's farm-to-table menu to deliver exceptional customer service.
- Enhanced teamwork and communication with colleagues to ensure smooth and efficient service operations.
- Created exceptional guest experiences through attentive service and a welcoming attitude.

Kiawah Island Golf Resort | *Server* | Kiawah Island, SC

May 2023 – May 2024

- Provided customers with extensive knowledge of the menu, showing strong interpersonal skills.
- Communicated effectively with staff, management, and the kitchen during service.
- Delivered memorable guest experiences through outstanding hospitality and efficient service.

Asheville Campus Entertainment | *Special Events Coordinator* | Asheville, NC

August 2022 – May 2023

- Orchestrated a diverse range of events to enhance the campus community experience.
- Coordinated with vendors, selected venues, sourced equipment, and managed schedules to ensure smooth execution.
- Engaged proactively with event-specific vendors to secure resources/services while negotiating favorable terms.

SKILLS

Software: SolidWorks CAD | Arduino | Vivado | Quartus | Vitis HLS | MATLAB/ Octave | UNIX/ Linux

Programming: Machine Code/ Binary | LC-3/ MIPS/ RISC-V | HTML/ CSS/ JS | Java | C/ C++/ C# | Python | R | VHDL/ Verilog